AFFORD-Bench: An Affordance-Grounded Benchmark and Generalization Evaluation for Robot Learning

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***Abstract*—Imitation learning policies and Vision-Language-Action (VLA) models report high success on standard manipulation benchmarks, yet LIBERO-PRO shows that these numbers can collapse when object positions or task goals change, which points to memorization of nominal scenes. Existing robustness suites perturb mostly tabletop pick-and-place scenes with centimeter-scale success predicates and rarely vary the quantities that set whether a contact-rich task can succeed: placement tolerance, friction, and mass. We introduce AFFORD-Bench, a policy-agnostic MuJoCo benchmark on a simulated Franka Emika Panda with 210 registered tasks across 7 domains. An audit of every task against its instruction yields a validated core of 66 pick, place, and insertion tasks in four domains; 35 further tasks are fixed-point reach tasks, and the remaining 109, including all multi-step chains, are disclosed as instruction mismatches. It ships 10,500 scripted-oracle demonstrations (1.23 M steps, dual-camera RGB) in Robomimic HDF5, four instruction variants per task, and a five-level perturbation hierarchy (L1–L5) over spatial, photometric, physical, and compound shifts. Grasping is frictional, and success is a physical predicate on the settled state: the released body must rest within 2–2.5 mm and 5° of its goal in the precision domains (8 mm and 15° elsewhere). A closed-loop privileged oracle solves 95.7% of nominal episodes and null and random policies solve none; the specified physical suite does not lower the oracle's success, but lower friction and heavier objects do (to 16%), which fixes its Stage 2 calibration.**

I. Introduction

Generalist Vision-Language-Action (VLA) models [11, 16, 17], continuous flow models such as π_0 [1], 3D spatial architectures such as SpatialVLA [19], visual reasoning policies such as CoT-VLA [30], compact VLAs such as SmolVLA [21], subtask-aware post-training such as AtomVLA [22], and action-chunking visuomotor policies [5, 31] all promise transfer across tasks and scenes.

Measuring that transfer remains difficult. LIBERO [13] is the de facto testbed, but Zhou et al. [32] (LIBERO-PRO) show that policies exceeding 90% nominal success can fall to near 0% when initial object positions or task goals change, and Wang et al. [27] (LIBERO-X) report an average success of 39.4% for five VLAs already at their mildest level of spatial shift. Both attribute the gap to memorized trajectories and layouts.

LIBERO-PRO, LIBERO-X, and LIBERO-Plus [6] perturb layout, appearance, camera, and language; COLOSSEUM [18] adds object friction and mass on the subset of its 20 RLBench tasks that support them. Their success predicates accept centimeter-scale placement errors (LIBERO's *On* allows 3 cm), and none of them couples these shifts to the quantity that decides success in precision work: how much positional error the task tolerates. In the sense of affordance theory [8], whether a part can be seated depends jointly on clearance, friction, and mass, not on appearance alone. Contact-rich assembly simulators such as Factory [15] and IndustReal [23] model these effects with high fidelity but target reinforcement learning on a few assemblies, not cross-domain evaluation of imitation and VLA policies. Requiring a placed body to be released and at rest is not new (ManiSkill3's stacking task does so [24]); we use it uniformly, at millimeter tolerances, across all evaluated tasks.

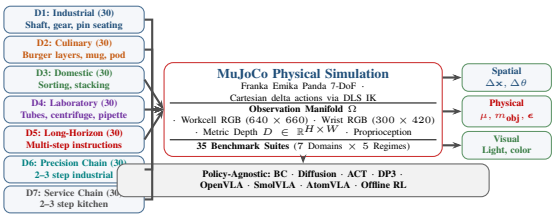


Fig. 1. **The AFFORD-Bench Architecture.** 210 registered tasks in 7 domains (66 audited core tasks in D1–D4), evaluated under spatial, physical, and visual shifts.

A benchmark should also be **policy-agnostic**: usable by behavior cloning, chunked diffusion and transformer policies (Diffusion Policy, ACT), point-cloud policies (DP3), VLAs (OpenVLA, SmolVLA, AtomVLA), and offline reinforcement learning (CQL, IQL).

AFFORD-Bench contributes:

1) **An audited task inventory**: 210 registered tasks, each checked at runtime against its instruction; **66 core tasks** in industrial seating, culinary stacking, tabletop sorting, and laboratory handling form the evaluated set, and every other task carries a stated reason for exclusion.
2) **Policy-agnostic data with integrity checks**: 10,500 oracle demonstrations with dual-camera RGB, proprioception, and full simulator states in Robomimic HDF5 [14], convertible to RLDS and LeRobot, checked episode by episode and by replay.
3) **Physical success at millimeter tolerance**: frictional, force-limited grasping with no attachment or goal snapping; success requires the released body to rest, supported, within 2–2.5 mm and 5° of its goal in the precision domains (8 mm and 15° elsewhere).
4) **Hierarchical perturbations**: a five-level hierarchy (L1

nominal, L2 spatial, L3 visual, L4 physical, L5 compound), with measured solvability floors for every level and task (Sec. IV).

5) **Multi-tier language**: four instruction variants per task (metric, contact-conditioned, terse, and subtask-decomposed with pre/post predicates), each annotated with whether the simulator evaluates what it asks.

II. Related Work

A. Robotic Manipulation Benchmarks

RLBench [10] and ManiSkill2 [9] provide simulated suites for reinforcement and demonstration-guided learning, ManiSkill2 with object-level geometric variation. Robomimic [14] standardized offline imitation learning pipelines across demonstration qualities. LIBERO [13] introduced 130 tasks for lifelong knowledge transfer. Apart from ManiSkill2's geometric variation, these suites evaluate mostly nominal execution without systematic out-of-distribution (OOD) testing.

B. OOD Robustness and Modern VLA Architectures

With large-scale pretrained policies [17, 3, 2, 11, 16], robustness to distribution shift has become a primary question. LIBERO-PRO [32] shows that policies with high nominal LIBERO success collapse when initial positions or task goals change, while remaining largely robust to object-appearance changes and paraphrases. LIBERO-X [27] adds hierarchical, cumulative perturbation levels with new human demonstrations, and LIBERO-Plus [6] analyzes seven perturbation factors over 10,030 task variants stratified into five difficulty levels. COLOSSEUM [18] varies 14 factors, including object friction and mass on the tasks that support them, and reports 30–50% degradation for single factors and at least 75% for combined ones. These evaluations remain concentrated on domestic tabletop scenes with centimeter-scale placement predicates.

C. Affordance and Contact-Rich Manipulation

Affordance theory [8] frames objects by the interactions they permit. In robotics this has motivated contact-rich grasp synthesis [26], spatial affordance maps [20], and 3D policy representations such as DP3 [29]. Factory [15] simulates contact-rich assembly fast enough for reinforcement learning, and IndustReal [23] transfers RL insertion and gear-meshing policies from such simulation to a real Franka. Diffusion Policy [5] and ACT [31] reach high precision on low-cost hardware [7], but how they degrade as tolerance tightens or friction changes is not characterized at scale. Offline RL [12] additionally needs transitions and rewards that VLA-specific datasets often omit. AFFORD-Bench targets this combination of millimeter-tolerance tasks (tolerance-indexed variants are a Stage 2 deliverable), multi-domain coverage, and policy-agnostic data.

III. The AFFORD-Bench Formulation

A. Tasks as Goal-Conditioned MDPs

Each task is a tuple $\kappa = (\ell, \mathbf{e}, \mathcal{C}_{\text{aff}})$ with instruction ℓ, environment $\mathbf{e} = (\mathcal{W}, \mathcal{O}, \rho_0, \Phi)$, and goal set $\mathcal{C}_{\text{aff}}$. $\mathcal{W}$ is the workspace geometry, $\mathcal{O} = \{o_1, \ldots, o_K\}$ the object set with one evaluated body $o^\star \in \mathcal{O}$, ρ_0 the initial-state distribution, and $\Phi = (\mu, m, \boldsymbol{\epsilon})$ the physical parameters: sliding friction μ of $o^\star$, its mass m, and success tolerance $\boldsymbol{\epsilon} = (\epsilon_{xy}, \epsilon_z)$.

Each task induces an MDP $\mathcal{M} = \langle \mathcal{S}, \mathcal{A}, P, R, \Omega, H \rangle$. $\mathcal{S}$ is the joint robot-object state. An action $\mathbf{a} = (\Delta\mathbf{p}, \Delta\boldsymbol{\theta}, a_{\text{grip}}) \in [-1, 1]^7$ commands a TCP position increment $0.025\,\Delta\mathbf{p}$ m, capped at 2.5 cm per step (0.5 m/s) and tracked by damped-least-squares IK at 20 Hz, a rotation increment $\Delta\boldsymbol{\theta}$ of which only the yaw component is executed ($0.05\,\Delta\theta_z$ rad per step, hand pointing down), and a per-finger opening target $w = 0.02\,(a_{\text{grip}} + 1)$ m, approached at no more than 0.05 m/s by a force-limited parallel jaw (about 35 N per finger). $P(s' \mid s, \mathbf{a})$ is given by MuJoCo [25] and is deterministic given s; all stochasticity enters through ρ_0 and the perturbation sampler, both seeded per episode. $R(s) = \mathbb{I}[s \in \mathcal{C}_{\text{aff}}]$ is sparse, and H is the step budget.

The goal set is a set of physical states:

$$\begin{aligned}\mathcal{C}_{\text{aff}} = \big\{ s \;\big|\; & \|\mathbf{p}^{xy}_{o^\star} - \mathbf{g}^{xy}\|_2 \le \epsilon_{xy},\ |p^z_{o^\star} - g^z| \le \epsilon_z, \\ & \theta_{o^\star} \le \epsilon_\theta,\ \mathcal{F}(o^\star) = \emptyset,\ \mathcal{K}(o^\star) \neq \emptyset, \\ & \|\mathbf{v}_{o^\star}\| < v_{\text{rest}},\ \|\boldsymbol{\omega}_{o^\star}\| < \omega_{\text{rest}} \big\}, \end{aligned} \tag{1}$$

where $\theta_{o^\star}$ is the tilt of $o^\star$ from upright, $\mathcal{F}$ and $\mathcal{K}$ are the finger bodies and the non-robot bodies in contact with $o^\star$, $v_{\text{rest}} = 0.02$ m/s, and $\omega_{\text{rest}} = 0.2$ rad/s. The goal $\mathbf{g}$ is the resting pose of $o^\star$ at the task's target, found when the environment is built by lowering $o^\star$ onto its support and simulating it to rest; there is no attachment, snapping, or teleport, so a body that slides, tips, or binds on a rim fails. Targets with no stable resting pose, such as holding a pipette above a stand, are hold goals, for which $\mathcal{F}(o^\star)$ must contain both fingers instead. Every manipulation task also requires $o^\star$ to have been lifted by $h_{\min} = 2$ cm (10–15 cm for four lift-and-replace tasks), so pushing $o^\star$ to the goal does not count. For reach-and-press tasks the condition is $\|\mathbf{p}_{\text{tcp}} - \mathbf{g}\|_2 \le \epsilon_{xy}$ for 5 consecutive steps. Episodes terminate at the first $t \le H$ with $s_t \in \mathcal{C}_{\text{aff}}$, so policy competence is

$$J(\pi; \kappa) = \mathbb{P}_{s_0 \sim \rho_0,\ \tau \sim \pi}\big[\exists\, t \le H : s_t \in \mathcal{C}_{\text{aff}}\big]. \tag{2}$$

Contact forces are monitored, not part of success: an episode violates safety if the net force $\mathbf{f}_c(s_t)$ that the scene, excluding $o^\star$, exerts on the robot exceeds $F_{\max} = 150$ N at some control step. In one oracle episode per task the peak force had median 0 N, 95th percentile 143 N, and maximum 169 N; 10 of 210 oracle episodes exceed $F_{\max}$, so we report the oracle's force violation rate (4.8%) as the reference level rather than treating violations as policy-specific.

B. Policy-Agnostic Benchmark Architecture

The observation $\mathbf{o}_t \in \Omega$ comprises **workcell RGB** ($H \times W = 640 \times 660$, uint8), **wrist RGB** (300×420, mounted at the gripper), optional **metric depth** (float32, back-projected as $\mathbf{P} = D(u, v)\,\mathbf{K}^{-1}[u, v, 1]^\top$ with the intrinsics $\mathbf{K}$ and camera poses stored in every dataset; not included in the released demonstrations), and **proprioception**: joint positions and

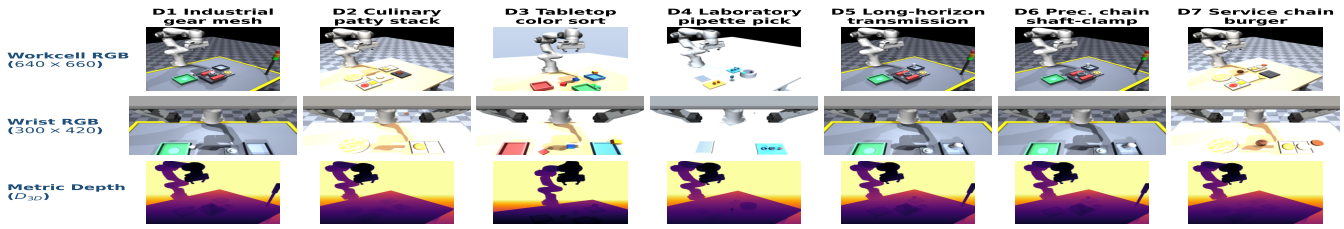


Fig. 2. **Multi-Domain Qualitative Visualizations in AFFORD-Bench.** Representative tasks from all seven domains: (Row 1) workcell RGB (640 × 660); (Row 2) wrist RGB (300 × 420); (Row 3) metric depth rendered from the workcell camera (meters).

velocities $\mathbf{q}, \dot{\mathbf{q}} \in \mathbb{R}^7$, TCP pose $\mathbf{x}_{\text{tcp}} \in \mathbb{R}^7$ (position and w-first quaternion), and per-finger opening $w \in [0, 0.04]$ m. Each step also records the sparse reward $R(s_t)$, the success flag, and the full simulator state.

C. Domain Architecture

As summarized in Table I and Figure 2, AFFORD-Bench has seven domains of 30 tasks each across five scenes: **D1: Industrial Precision** (001–030): shaft, gear, and pin seating in a jig, kitting, and fixture contact. **D2: Culinary Operations** (031–060): burger layer stacking, mug and capsule handling at an espresso station. **D3: Domestic Tabletop** (061–090): color sorting into bins, stacking, and relocation. **D4: Laboratory Sciences** (091–120): tube racking, centrifuge loading, and pipette positioning. **D5–D7** (121–210): instructions that chain 2–4 steps across industrial, laboratory, and kitchen scenes. In Stage 1 each of these tasks evaluates a single body, and the episode ends when that body reaches its goal, which in 57 of 90 tasks occurs before the last instructed step. Multi-body sequential predicates that score every step are a Stage 2 deliverable.

Task audit and tiers. Every task was rebuilt and reset at runtime to record the body under its goal, whether the goal is a resting or a hold goal, and the spread of its starts. A task is *core* if its instruction names the simulated body and its success predicate covers the whole instruction (66 tasks). The 35 *reach* tasks require the fingertip center to stay within tolerance of a fixed world point for 5 steps; their instructions mention wiping, pressing, or inspecting, none of which is simulated, so they are reported separately. The other 109 are *instruction mismatches*: 66 name an object that is not the simulated body (e.g., a "bearing ring" that is the spur gear), 31 are chains whose single-body predicate covers only part of the instruction, and 12 contradict the predicate in other ways (e.g., "hold the pipette tip over the plate" where success requires releasing the pipette onto it). In 17 core tasks the goal is an unmarked point on a table, bench, or board, and in 13 the start varies by less than 1 cm because the body starts in a tray or on a spindle; we report both groups separately. Results in this paper are for all 210 tasks unless marked as core.

IV. The Hierarchical Affordance Perturbation Framework

AFFORD-Bench organizes distribution shift into five levels:

L1: Nominal (in-distribution). The start of $o^\star$ is drawn from $\mathcal{U}(\pm 7\,\text{cm})$ per horizontal axis around its nominal position, yaw from $\mathcal{U}(-45°, 45°)$, and roll and pitch from $\mathcal{U}(-10°, 10°)$ at spawn, resolved by simulating the body to rest; the initial TCP is jittered by ± 1.5 cm per horizontal axis. Starts are rejection-sampled: $o^\star$ must rest flat on a single support (tilt below 3°, support height spread at most 2 mm), have no fixture overhead, admit a collision-free top-down parallel-jaw grasp, and lie away from the goal. If no valid start is found, the scatter is shrunk to $\frac{1}{2}$, $\frac{1}{4}$, $\frac{1}{10}$, and $\frac{1}{20}$ before falling back to the nominal pose, so bodies that start in a compartment or seated on a shaft vary much less than the nominal range: over 10 audit resets, the start varies by less than 1 cm (standard deviation) in 29 of 210 tasks, and task 12 always starts at its nominal pose. D1, D3, and D7 scenes add two distractor bodies of similar color or shape, sampled at least 9 cm from the start and goal; after settling, a distractor ends closer than 9 cm in 7 of 76 tasks (minimum 3.9 cm).

L2: Spatial ($\mathcal{P}_{\text{spatial}}$). The start offset has radius $\mathcal{U}(7, 12)$ cm in a uniformly random direction, outside the L1 support, and yaw magnitude $\mathcal{U}(45°, 90°)$, probing closed-loop servoing versus trajectory recall.

L3: Visual ($\mathcal{P}_{\text{visual}}$). Each scene light's diffuse intensity is scaled by $\mathcal{U}(0.5, 1.5)$ and clipped at full intensity (so the key light, nominally 0.9, brightens by at most $1.11\times$), light positions are jittered by ± 0.5 m, and object color by $\pm 25\%$, leaving geometry unchanged.

L4: Physical ($\mathcal{P}_{\text{physics}}$). Sliding friction $\mu \sim \mathcal{U}(0.12, 0.25)$ on $o^\star$ and on both finger pads, and mass and rotational inertia of $o^\star$ scaled by $\mathcal{U}(1.8, 2.5)$, at identical visual coordinates. MuJoCo takes the larger friction of a contact pair, so pads and body are set together. With the elliptic friction cones used here, this range does not make the tasks harder for the privileged oracle, which centers its grasps: it solves 95.0% of L4 episodes against 95.7% of nominal ones on the same seeds (Table II). In a sweep over the 66 core tasks (3 seeds per setting), oracle success stays at or above 94% for $\mu \geq 0.12$ up to $2.5\times$ mass, but falls to 70% at $\mu = 0.12$ and $5\times$ mass, 38% at $\mu = 0.05$ and $5\times$, and 16% at $\mu = 0.02$ and $5\times$. The physical axis therefore matters, but outside the range specified here; recalibrating L4 on this curve is part of the Stage 2 charter.

L5: Compound ($\mathcal{P}_{\text{comp}}$). L2, L3, L4, and color shifts sampled jointly, combined with the multi-step instructions

TABLE I

TAXONOMY OF THE AFFORD-BENCH SUITE. 210 REGISTERED TASKS IN 7 DOMAINS. TOLERANCES ARE THE SUCCESS THRESHOLDS $(\epsilon_{xy}, \epsilon_z, \epsilon_\theta)$; D5 TASKS TAKE THE TIER OF THEIR SCENE. D1, D3, AND D7 SCENES ADD TWO DISTRACTOR BODIES; TOLERANCE-INDEXED VARIANTS ARE A STAGE 2 DELIVERABLE. AUDITED TIERS (CORE / REACH / INSTRUCTION MISMATCH): D1 17/6/7, D2 20/7/3, D3 17/8/5, D4 12/14/4, D5–D7 0/0/30 EACH.

Domain	Task IDs	Count	Representative Manipulation Tasks	Evaluated Bodies	Tol. (mm / deg)
D1: Industrial Precision	001–030	30	Shaft-in-bore, gear-on-shaft, pin-in-hole, kitting, fixture contact	Shaft, spur gear, dowel pin	2 / 5
D2: Culinary Operations	031–060	30	Burger layer stacking, mug to drip tray, capsule to hopper	Patty, cheese, tomato, buns, mug, capsule	8 / 15
D3: Domestic Tabletop	061–090	30	Color sorting, block stacking, relocation, touch	Two cubes, one cylinder	8 / 15
D4: Laboratory Sciences	091–120	30	Tube racking, centrifuge loading, pipette positioning	Conical tube, microtube, pipette	2 / 5
D5: Long-Horizon Sequences	121–150	30	Multi-step assembly, sorting, and service instructions	One per task (Stage 1)	2 or 8 / 5 or 15
D6: Precision Chaining	151–180	30	2–3 step industrial and laboratory chains	One per task (Stage 1)	2.5 / 5
D7: Service Assembly	181–210	30	2–3 step kitchen and household chains	One per task (Stage 1)	8 / 15
Total	**001–210**	**210**	**7 Domains × 30 Tasks (35 Evaluation Suites)**	**16 rigid bodies**	**2–8 / 5–15**

of D5–D7.

Physics and implementation. All five levels are implemented. Lights, colors, friction, and masses are restored to nominal before each sample, and all sampling is seeded per episode. Grasping is frictional: the arm is driven by gravity-compensated joint position servos, the gripper is a force-limited tendon actuator, finger pads use stiff contacts (4 ms time constant) with elliptic friction cones (impratio $= 10$), and collisions use MuJoCo's native convex collision with multi-point contact. At reset, bodies are lifted out of any interpenetration and simulated to rest before the first observation.

V. ALGORITHMIC ORACLES AND DATASET SYNTHESIS

A. Closed-Loop Oracles

Teleoperated demonstrations are costly and inconsistent [14]. AFFORD-Bench instead uses a single privileged, closed-loop oracle for all 210 tasks that reads the pose of $o^\star$ at every step and acts through the same controller and action space as learned policies. It plans a grasp by checking candidate yaws, grasp heights, and finger pre-shapes with inverse kinematics and collision detection on a scratch copy of the state, approaches with a trapezoidal velocity profile (1.6 m/s^2), closes and verifies a two-finger grasp (up to three attempts), carries $o^\star$ at reduced speed and acceleration (0.24 m/s, 0.8 m/s^2) above every surface on the straight path to the goal while correcting for its offset in the hand, rotates the hand if the fingers would collide at the goal, lowers and re-centers $o^\star$ until it is supported, lifts and re-centers after a stalled insertion (up to three times), waits for $o^\star$ to come to rest, and opens until neither finger touches it before retracting. The oracle uses state that policies do not observe, so its success bounds solvability rather than predicting learned performance.

B. Demonstration Scale and Packaging

The released dataset holds 50 successful oracle demonstrations for each of the 210 tasks (10,500 in total, 1,230,842 steps, mean 117 steps per demonstration) in Robomimic HDF5 with 90%/10% train/validation masks, the full simulator state at every step, the reset seed, and camera intrinsics and poses. Each demonstration starts from its own seed (all initial states are distinct), but only successful oracle episodes are kept: about 5% of sampled starts were discarded, so the training start distribution is conditioned on oracle success. An episode-level validator checks every demonstration (shapes, finite values, terminal reward and done, alignment of observations with recorded states, duplicates, splits) and finds no errors; replaying the recorded actions of 630 sampled demonstrations reproduces the exact success step in 595 (median deviation of $o^\star$ 0.02 mm); the other 35 stem from storing float32 roundings of the executed actions, which the generator now avoids (its episodes replay bit-identically). Known limitations: 114 demonstrations run longer than the environment's 500-step limit, the state after the final action is not stored, and the files were produced by three generator invocations whose code version was not recorded; the released generator now records provenance and rejects over-length demonstrations. The converters of Sec. VIII export the data to RLDS (OpenVLA, Octo) and LeRobot (SmolVLA).

VI. TRAINING AND EVALUATION PROTOCOLS

A. Standardized Closed-Loop Rollout Procedure

Each policy step applies one action at 20 Hz: the arm is interpolated to the IK joint target over 25 physics steps of 2 ms, with commanded TCP speed capped at 0.5 m/s. For each task we roll out $N = 50$ episodes with evaluation seeds disjoint from data-generation seeds, with an evaluation step budget of $H = 400$ steps (environment step cap 500; Stage 2 multi-body chains extend H to 800). Every suite, including nominal, draws its initial state from the evaluation seed.

B. Standardized Evaluation Suites

Models are evaluated on 35 suites (7 domains × 5 conditions): (1) **Nominal**; (2) **Spatial** ($\mathcal{P}_{\text{spatial}}$); (3) **Visual** ($\mathcal{P}_{\text{visual}}$); (4) **Physical** ($\mathcal{P}_{\text{physics}}$); and (5) **Composite** ($\mathcal{P}_{\text{comp}}$), with parameters as in Sec. IV. Language variants are evaluated as a separate axis on the nominal scene.

C. Quantitative Evaluation Metrics

We report: (i) **Success Rate** $\text{SR} = \frac{1}{N}\sum_{i=1}^{N} \mathbb{I}[\exists\, t \le H : s_t^{(i)} \in \mathcal{C}_{\text{aff}}]$ with Wilson 95% intervals over episodes and mean ± s.d. over 3 training seeds; (ii) **Physical Robustness Index** $\text{PRI}(\mathcal{P}) = \text{SR}(\mathcal{P})/\text{SR}(\text{Nominal}) \times 100\%$, defined when $\text{SR}(\text{Nominal}) > 0$; (iii) **Generalization Degradation Index** $\text{GDI} = 100\% - \text{PRI}$; and (iv) **Force Violation Rate** (FVR), the fraction of episodes violating $F_{\max}$.

TABLE II

STAGE 1 CALIBRATION. SUCCESS (%), MUJOCO 3.5.0. ORACLE: 5 EPISODES PER TASK AND SUITE ON SHARED SEEDS, WITHIN ITS 600-STEP BUDGET OR WITHIN $H = 400$. RANDOM (5 PER TASK) AND NULL (1 PER TASK) ON L1, $H = 400$. BRACKETS: 95% TASK-BOOTSTRAP INTERVALS.

Task Domain	Oracle	Oracle (H)	Oracle (L4)	Random	Null
D1: Industrial Precision	95.3	95.3	96.7	0.0	0.0
D2: Culinary Operations	98.0	97.3	99.3	0.0	0.0
D3: Domestic Tabletop	100.0	100.0	100.0	0.0	0.0
D4: Laboratory Sciences	93.3	90.0	90.7	0.0	0.0
D5: Long-Horizon Sequences	93.3	93.3	92.0	0.0	0.0
D6: Precision Chaining	92.0	92.0	90.0	0.0	0.0
D7: Service Assembly	98.0	93.3	96.0	0.0	0.0
Core tier (66 tasks)	95.2 [90.9, 98.5]	93.3	96.1 [92.4, 99.1]	0.0	0.0
All 210 tasks	**95.7** [93.8, 97.4]	**94.5**	**95.0** [93.0, 96.8]	**0.0**	**0.0**

VII. SCIENTIFIC HYPOTHESES & PRELIMINARY CALIBRATION

A. Core Scientific Hypotheses

AFFORD-Bench is designed to test three hypotheses; H_2 requires tolerance-indexed variants and H_3 multi-body predicates (both Stage 2):

- H_1 **(physical versus appearance shift)**: *policies with high nominal success lose more under $\mathcal{P}_{\text{physics}}$ than under $\mathcal{P}_{\text{visual}}$ of matched severity, with object 2D coordinates unchanged.*
- H_2 **(tolerance fragility)**: *at fixed spatial shift, success decreases monotonically as ϵ_{xy} shrinks; tested with tolerance-indexed variants of D1 and D4 tasks at $\epsilon_{xy} \in \{2, 4, 8, 16\}$ mm.*
- H_3 **(error compounding)**: *in D6 and D7, the success probability of a k-step chain falls below the product of its per-step success rates measured in isolation.*

B. Preliminary Empirical Calibration

Table II reports solvability under the protocol of Secs. III and IV on the released code and simulator. The oracle solves 1005 of 1050 nominal episodes (95.7%); 28 tasks fail at least once, concentrated in a few (`lab_micro_tube_to_rack_b` fails all 5 episodes, `chain_sensor_bracket_mount` 4 of 5). Within the evaluation budget of 400 steps the rate is 94.5%, because some chained tasks need up to 600 steps. Random actions solve none of 1050 episodes and the null policy none of 210; the lift requirement and the release-and-rest predicate remove trivially satisfied goals. Under L4 the oracle succeeds in 95.0% of episodes, so at the specified ranges the physical suite does not lower expert solvability (Sec. IV); H_1 therefore has to be tested on learned policies, with an L4 range recalibrated on the friction and mass sweep of Sec. IV. As a check that the demonstrations are learnable, a behavior-cloning MLP given proprioception and the pose of $o^\star$ (trained 200 epochs on the 45 training demonstrations of task 61) solved 16 of 20 evaluation episodes; this is a pipeline check, not a baseline result. All numbers use one simulator build (MuJoCo 3.5.0) and MuJoCo Menagerie meshes for the Panda [28].

VIII. DISCUSSION AND STAGE 2 EXTENSION CHARTER ROADMAP

A. Physical Hardware Correspondence and Sim-to-Real Feasibility

Controlled comparison needs what hardware cannot provide: exact resets, perturbations sampled from known distributions, and ground-truth success labels. Fifty episodes per task and condition over the 66 core tasks and five suites are 16,500 rollouts per policy, each replayable from its seed, or about 140 robot-hours at two trials per minute, over which wear and lighting drift would confound the comparison. We therefore use simulation as the instrument and hardware to test whether its rankings hold. The modalities (fixed and wrist RGB, proprioception) match Franka workcells with RealSense cameras and LeRobot [4] setups, and `scripts/convert_to_lerobot.py` and `scripts/convert_to_rlds.py` export LeRobot v2.0 and the RLDS layout of Open X-Embodiment [17] (round-trip tests on synthetic episodes; RLDS in its NumPy layout, not through TFDS). Three gaps remain: image sizes differ from native RealSense modes; the action is a Cartesian increment, whereas SO-101, Piper, and ALOHA arms are commanded in joint space; and bodies and fingertips are rigid, with contact softness set by the solver.

The charter's hardware pilot runs two D1 core tasks (peg insertion, Task 003; gear meshing, Task 002) on a Franka Panda with wrist and fixed RealSense cameras and 3D-printed replicas at three clearances. Policies trained on simulated, teleoperated, and pooled demonstrations are evaluated under nominal, spatial, and physical (friction films, added mass) conditions, 20 trials each, and sim-to-real agreement is the rank correlation of per-condition success. Unlike IndustReal [23], the question is whether simulated perturbation rankings predict physical degradation for imitation and VLA policies, not whether one policy transfers.

B. Stage 2 Extension Charter

In accordance with the RSS 2027 shepherding process, the Stage 2 charter comprises: (1) *Simulator fidelity*: tolerance-indexed task variants, multi-body predicates for D5–D7, compliant fingertip models, and an L4 range calibrated against an oracle-degradation curve over friction and mass; (2) *Full empirical scaling*: 50 rollouts per condition on the 66 core tasks and all five suites for reference implementations of Diffusion Policy and ACT and for OpenVLA, SmolVLA, AtomVLA, π_0, SpatialVLA, CoT-VLA, and DP3, with three training seeds each; (3) *Ablations*: sensitivity curves for each perturbation and for tolerance; (4) *Hardware pilot* as above; and (5) *Domain 8 (Tasks 211–240)*: tool use, non-prehensile manipulation, twist-lock assembly, and mid-trajectory perturbations, admitted only at $\geq 95\%$ oracle and 0% null and random success. A 4 cm mid-approach displacement already separates closed-loop and 16-step chunked execution of one feedback law (14 of 18 episodes each) from open-loop replay (0 of 18).

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